CartPole Q-Learning Implementation

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CartPole can be modeled as a Markov decision process (MDP)

$$(\mathcal{S}, \mathcal{A}, P, R, \gamma),$$

where the state is typically

$$s_t = (x_t, \dot{x}_t, \theta_t, \dot{\theta}_t),$$

with x_t the cart position, $\dot{x}_t$ the cart velocity, θ_t the pole angle, and $\dot{\theta}_t$ the pole angular velocity.

The action space is discrete:

$$\mathcal{A} = \{\text{left, right}\},$$

which can also be written as $a_t \in \{-1, +1\}$ when using a signed force convention. At each time step, the agent applies a force F_t to the cart, and the environment evolves according to the following nonlinear dynamics of the cart-pole system:

$$\begin{aligned}
\ddot{x}_t &= \frac{F_t + m_p \sin(\theta_t)(l\dot{\theta}_t^2 + g\cos(\theta_t))}{m_c + m_p \sin^2(\theta_t)} \\
\ddot{\theta}_t &= \frac{-F_t \cos(\theta_t) - m_p l \dot{\theta}_t^2 \sin(\theta_t)\cos(\theta_t) - (m_c + m_p) g \sin(\theta_t)}{l(m_c + m_p \sin^2(\theta_t))},
\end{aligned}$$

where m_c is the mass of the cart, m_p is the mass of the pole, l is the length of the pole, and g is the acceleration due to gravity.

The control objective is to keep the pole balanced near the upright position $\theta = 0$ while keeping the cart near the track center. A standard reward choice is

$$r_t = 1$$

for every step before termination. An episode ends when the cart leaves the allowed position range or when the pole angle exceeds a threshold.

For Q-learning, the action-value function $Q(s, a)$ is updated by

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha\Big(r_t + \gamma \max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t)\Big),$$

where α is the learning rate and γ is the discount factor.